Effect of diet and exercise on serum insulin, IGF-I, and IGFBP-1 levels and growth of LNCaP cells in vitro (United States).

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OBJECTIVE: Accumulating evidence indicates that prostate cancer is associated with high levels of serum IGF-I. This study was conducted to determine whether a low-fat diet and exercise (DE) intervention may modulate the IGF axis and reduce prostate cancer cell growth in vitro. METHODS: Fasting serum was obtained from 14 men (age 60 +/- 3 years) participating in an 11-day DE program and from eight similarly aged men who had followed the DE program for 14.2 +/- 1.7 years (long-term). Insulin, IGF-I, IGFBP-1, and IGFBP-3 were measured by ELISA, and serum was used to stimulate LNCaP cell growth in vitro. RESULTS: Serum IGF-I levels decreased by 20% while IGFBP-1 increased by 53% after 11-day DE. In the long-term group, IGF-I was 55% lower, while IGFBP-1 was 150% higher relative to baseline. Serum insulin decreased by 25% after 11-day DE and was 68% lower in the long-term group, relative to baseline. No changes in serum IGFBP-3 were observed. Serum-stimulated LNCaP cell growth was reduced by 30% in post-11-day serum and by 44% in long-term serum relative to baseline. LNCaP cells incubated with post-DE serum showed increased apoptosis/ necrosis, compared to baseline. CONCLUSIONS: A low-fat diet and exercise intervention induces in-vivo changes in the circulating IGF axis and is associated with reduced growth and enhanced apoptosis/necrosis of LNCaP tumor cells in vitro.